

1. Title

Predicting Student Dropout and Academic Success in Higher Education with Supervised Classification Models

2. Abstract

This project proposes a supervised machine-learning study that predicts whether a university student will drop out, stay enrolled, or graduate. The team will use the Predict Students' Dropout and Academic Success dataset from the UCI Machine Learning Repository, which contains 4,424 student records from a Portuguese higher-education institution. Each record has 36 fields such as age at enrollment, course, admission grade, scholarship and tuition status, and first and second semester course results.

The central research question is how accurately these three outcomes can be predicted, and how early. In particular, the team wants to know how much accuracy is lost when a model only uses information available at enrollment and after the first semester, instead of the full two semesters of results. The study will begin with data inspection and a reproducible train/validation/test split.

A majority-class benchmark and a logistic regression baseline will be compared with two stronger models, a decision tree and a random forest. Performance will be evaluated primarily with macro-averaged F1, supported by accuracy and per-class precision and recall. The expected contribution is a transparent proof of concept that shows whether dropout can be flagged early from administrative records, and where the model is less reliable.

3. Research Background

Dropout is costly for students and for institutions, and it is much easier to help a student who is identified early than one who has already left. Research on student retention goes back decades [1], but in practice advisors often notice a struggling student only after grades or tuition payments have already gone wrong. This makes the problem a good fit for supervised classification: the model learns from past students whose outcomes are known, then predicts the outcome for new students.

The dataset used here was built by Realinho et al. from several databases of a Portuguese institution, and their work showed that early academic performance, together with financial and background variables, carries real predictive signals [2][3]. Many public notebooks on this data report accuracy in the mid-70s percent using all 36 features. Two things are often missing from that work. Firstly, the second-semester results are included even though they arrive too late for early help. Secondly, the smaller Enrolled class (about 18% of students) is rarely examined, and it seems tuning and test-set use are not always documented.

A key limitation we noticed is that the data comes from a single institution and are record based, so the project will not make claims about what causes a student to drop out. The goal is quite predictive and comparative, which is to see how well accessible records can flag at risk students, and to be honest about

where the model fails. Because the data include attributes such as gender, age, and nationality, the team will also check whether errors are uneven across groups.

- RQ1. How much better than naive benchmarks can logistic regression, a decision tree and a random forest predict the three outcomes, measured by macro-F1?
- RQ2. How much predictive performance is lost when only enrollment and first-semester information (30 features) is used instead of all 36 features?
- RQ3. Is performance uneven across subgroups (gender, scholarship holders, students aged 25 or older), and do the financial features (tuition up to date, debtor) drive the results?

4. Methodology

Data: The primary data source is the UCI dataset above (4,424 rows, 36 features plus a three-class target: Dropout, Enrolled, Graduate). The unit of analysis is an individual student. The classes are a little bit imbalanced (Graduate 49.9%, Dropout 32.1%, Enrolled 17.9%), and the file has no missing values. The dataset is anonymized and public and its license and source will be documented in the Milestone 2 data card.

Preprocessing and features: Integer-coded categorical variables such as course and application mode will be one-hot encoded and numeric variables standardized, all inside a pipeline fit on training data only so nothing leaks from the validation or test sets. Two feature sets will be compared: an early set (enrollment information plus first-semester results, 30 features) and a full set (adds second-semester 2 results, 36 features). Unusual cases, such as 180 students with zero enrolled units, will be examined before modelling, and financial variables such as tuition status will be checked for possible timing problems.

Models and comparison: The project will begin with a majority-class benchmark and a multinomial logistic regression baseline. Two further models, a decision tree and a random forest, will be compared against them. All models will use the same stratified 70/15/15 split and preprocessing. Hyperparameters will be chosen with cross-validation on the training data, with the final test set used only once at the end.

Evaluation: Macro-F1 will be the primary metric, because accuracy can look good while the minority classes are ignored (always predicting Graduate gets about 50% accuracy). Accuracy, per-class precision and recall, and confusion matrices will also be reported, along with variability across folds or random seeds. The team will compare early versus full features, test performance without the financial variables, and inspect errors by group (gender, scholarship status, age at enrollment). As a success criterion set before the final experiments, the best model should beat the best naive benchmark by at least 0.30 macro-F1 using the early feature set. Negative or inconclusive results will be reported as they are.

5. Significance

A reliable early warning model has great potential to help advisors and offices for student support to decide who to contact first, and show how much accuracy depends on information that only arrives late in the year. The project can also be useful for the course because it connects core Machine Learning I topics

such as classification, class imbalance, feature engineering, validation, baseline comparison, and error analysis to a real tabular dataset.

The project will not replace human advising or prove why students leave. However, it can produce an evidence-based pipeline and a careful comparison of an interpretable model and a more flexible one. That contribution can support later work on probability calibration, fairness analysis, or testing on data from other institutions.

6. SMART Timeline

Week 1 (by Oct 2)

Finalize the research question and dataset, write the data card, complete the initial analysis, fix the split and primary metric, and submit the Milestone 2 approval memo.

Week 2

Build the preprocessing pipeline, the stratified train/validation/test split, categorical encoding, and the early and full feature sets.

Week 3 (by Oct 19)

Implement the majority-class benchmark and logistic regression baseline, record initial macro-F1 and accuracy in a reproducible notebook or script, and submit Milestone 3.

Week 4 (by Oct 30)

Train the decision tree and random forest on the same split, prepare the theory explanation (softmax and cross-entropy loss, Gini impurity), and give the Milestone 4 progress presentation.

Week 5

Tune the models with cross-validation using comparable search effort, and log every run, including unsuccessful ones.

Week 6 (by Nov 20)

Run the early versus full comparison, the version without financial variables, the group error analysis, and the error analysis of the Enrolled class; submit Milestone 5.

Week 7

Fix remaining limitations, finalize tables and figures, and draft the final report.

Final week (by Dec 4)

Write the final report, prepare the presentation, make sure the repository reproduces the main results, and complete the contribution and AI-use documentation. Individual reflections follow on Dec 7.

7. References

1. V. Tinto, "Dropout from higher education: A theoretical synthesis of recent research," *Review of Educational Research*, vol. 45, no. 1, pp. 89-125, 1975.
2. V. Realinho, J. Machado, L. Baptista, and M. V. Martins, "Predicting student dropout and academic success," *Data*, vol. 7, no. 11, p. 146, 2022, doi: 10.3390/data7110146.
3. V. Realinho, M. V. Martins, J. Machado, and L. Baptista, "Predict Students' Dropout and Academic Success," *UCI Machine Learning Repository*, 2021, doi: 10.24432/C5MC89.
4. F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825-2830, 2011.